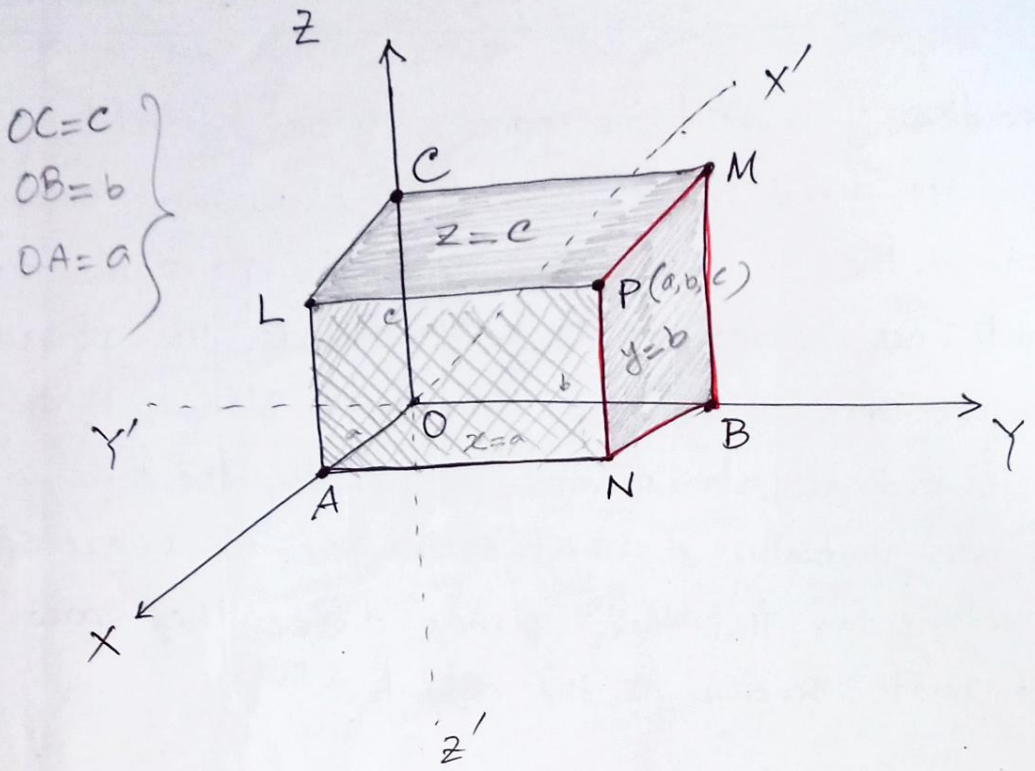
Three-dimensional coordinate system

The position of a point in space may be defined by referring it to three fixed planes. These planes have a common point, known as the origin and intersect in pairs in three straight lines, known as Co-ordinate Axes. The plane themselves are called the co-ordinate planes. When the three co-ordinate planes and consequently the three co-ordinate axes are mutually at right angles, the axes are said to be rectangular. The three planes divide the space into eight parts known as the octants.

Rectangular coordinate system in 3-space:

Let XOX', YOY', and ZOZ' be three fixed intersecting lines which are not coplanar and are mutually perpendicular to each other. These three lines are called the coordinate axes. More precisely, the lines $X'OX$, $Y'OY$, and $Z'OZ$ are called the x-axis, the y-axis, and the z-axis, respectively. The point O which is the point of intersection of these coordinate axes is called the origin. Let the positive directions of the axes are chosen to be $X'OX$, $Y'OY$, and $Z'OZ$.

Let P be any point in space. Through P the planes PNAL, PNBM, and PLCM are drawn parallel to the planes YOZ, ZOX, and XOY, respectively cutting $X'X$, $Y'Y$, and $Z'Z$ in A, B, and C, respectively.



The position of the point P relative to the axes can be determined if the magnitudes and signs of OA, OB, and OC are known. The length of the segment OA measured from O to A are taken with its proper sign is denoted by x and is called the x-coordinate of the point P. Similarly the measures of OB and OC with their proper signs are called the y-coordinate and z-coordinate respectively of the point P. The three numbers x, y, z which thus define the position of the point P in space are called the coordinates of the point P relative to the axes and the point P, in this sense, is represented as P(x, y, z).

P(a, b, c)

The coordinate planes divide the space into eight parts

Called the octants and the signs of the coordinates of a point are determined by the octant in which it lies. The table below gives the signs for the eight octants.

Octants	$OXYZ$	$OXYZ'$	$OXY'Z$	$OX'YZ$	$OXY'Z'$	$OX'YZ'$	$OX'Y'Z$	$OX'Y'Z'$
x	+	+	+	−	+	−	−	−
y	+	+	−	+	−	+	−	−
z	+	−	+	+	−	−	+	−

Notes:

i) The origin has the coordinates $(0,0,0)$

ii) Any point on the x-axis has the coordinates $(x,0,0)$

iii) Any point on the y-axis has the coordinates $(0,y,0)$

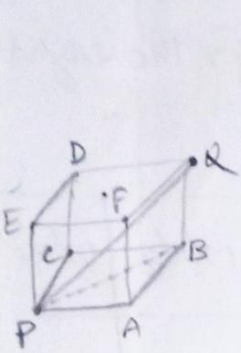
iv) Any point on the z-axis has the coordinates $(0,0,z)$

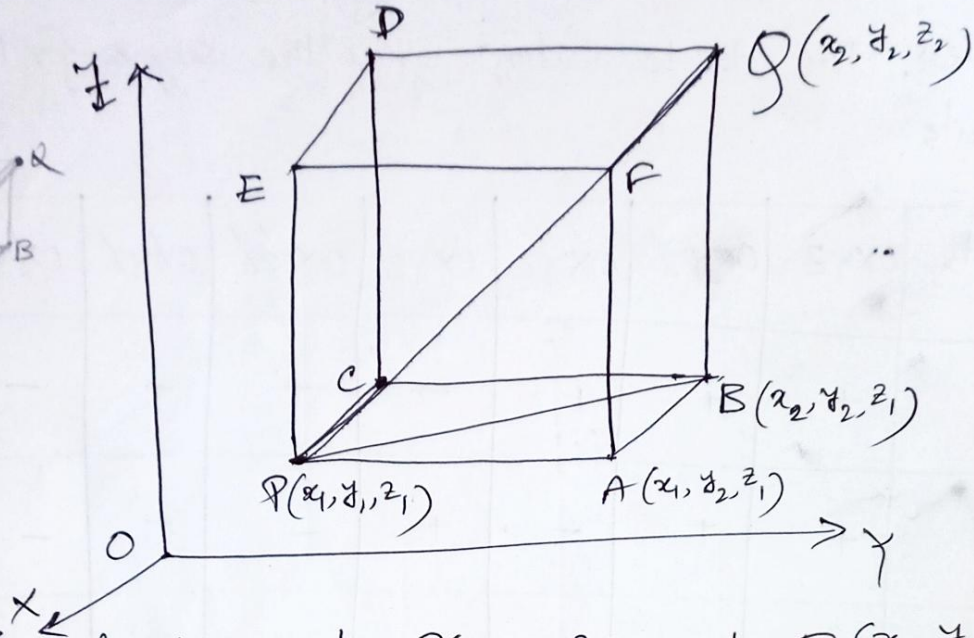
v) On the YOZ-plane, $x=0$ everywhere

vi) On the ZOX-plane, $y=0$ everywhere

vii) On the XOY-plane, $z=0$ everywhere

Distance between two points:





Through the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ draw planes parallel to the co-ordinate planes to form a rectangular parallelopiped whose one diagonal is PQ. Then PAFE, CBQD, PABC, EFQD, PCDE and ABQF are the three pairs of parallel faces of this parallelopiped.

Now, ΔPAB is a right-angled triangle where $\angle PAB = 90^\circ$

$$\therefore PB^2 = PA^2 + AB^2 \quad \text{——— (1)}$$

From the figure, $PA = y_2 - y_1$

$AB = x_2 - x_1$

& $BQ = z_2 - z_1$ ——— (2)

Therefore,

$$PB^2 = (y_2 - y_1)^2 + (x_2 - x_1)^2 \quad \text{——— (3)}$$

Also, ΔPBQ is a right-angled triangle with $\angle PBQ = 90°$

$$\therefore PQ^2 = PB^2 + BQ^2$$

$$\Rightarrow PQ^2 = (y_2 - y_1)^2 + (x_2 - x_1)^2 + (z_2 - z_1)^2 \quad [\text{From } (2) \text{ \& } (3)]$$

$$\Rightarrow PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Hence, the distance betⁿ two points $P(x_1, y_1, z_1)$ & $Q(x_2, y_2, z_2)$ is given by.

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad \text{——— (4)}$$

Distance from the origin :

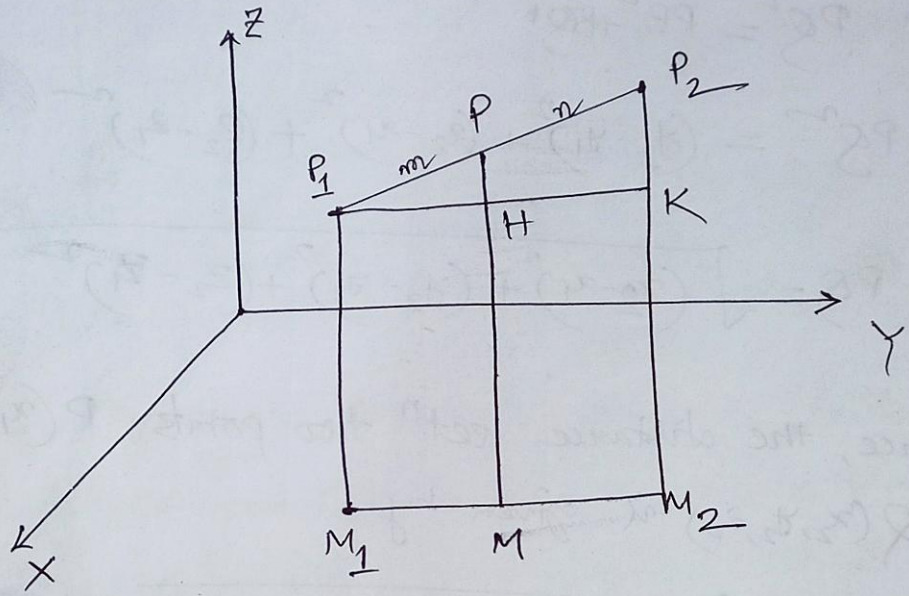
When the point $P(x_1, y_1, z_1)$ coincides with the origin $O(0,0,0)$, then we have $x_1 = 0$, $y_1 = 0$, $z_1 = 0$

Thus, from eqⁿ (4) we can write,

$$PQ = \sqrt{x_2^2 + y_2^2 + z_2^2}.$$

Section formula :

To find the coordinates of the point $P(x, y, z)$ which divides the straight line joining the points $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ in a given ratio $m:n$ internally.



Let $P(x, y, z)$ be the point which divides the straight line joining the points $P_1(x_1, y_1, z_1)$ & $P_2(x_2, y_2, z_2)$ in the ratio $m:n$ internally.

Draw P_1M_1, PM, P_2M_2 perpendiculars on the XOY plane. Clearly M_1, M_2, M are collinear and P_1M_1, P_2M_2 and PM are coplanar. Draw P_1K parallel to M_1M_2 meeting PM, P_2M_2 in H and K, respectively.

Then, $P_1M_1 = z_1$, $PM = z$ and $P_2M_2 = z_2$

$\therefore PH = PM - HM$

$= PM - P_1M_1$

$= z - z_1$

$P_2K = P_2M_2 - KM_2$

$= P_2M_2 - P_1M_1$

$= z_2 - z_1$

From the similar triangle P_1HP & P_1KP_2 we have

$$\frac{P_1P}{P_1P_2} = \frac{PH}{P_2K}$$

$$\Rightarrow \frac{P_1P}{P_1P_2 - P_1P} = \frac{PH}{P_2K - PH}$$

$$\Rightarrow \frac{P_1P}{PP_2} = \frac{PH}{P_2K - PH}$$

$$\Rightarrow \frac{m}{n} = \frac{z - z_1}{(z_2 - z_1) - (z - z_1)}$$

$$\Rightarrow \frac{m}{n} = \frac{z - z_1}{z_2 - z_1 - z + z_1}$$

$$\Rightarrow \frac{m}{n} = \frac{z - z_1}{z_2 - z} \Rightarrow m(z_2 - z) = n(z - z_1)$$

$$\Rightarrow mz_2 - mz = nz - nz_1$$

$$\Rightarrow (m+n)z = mz_2 + nz_1$$

$$\therefore z = \frac{mz_2 + nz_1}{m+n}$$

Similarly,

$$x = \frac{mx_2 + nx_1}{m+n}$$

&

$$y = \frac{my_2 + ny_1}{m+n}$$

$$P(x, y, z) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n} \right)$$

When P divides the straight line P_1P_2 externally then we have

$$x = \frac{mx_2 - nx_1}{m-n}$$

$$y = \frac{my_2 - ny_1}{m-n}$$

$$z = \frac{mz_2 - nz_1}{m-n}$$

If P is the mid-point of P_1P_2 then

$$P\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2} \right)$$